Addis Ababa University
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Department of Physics

Computational Physics Application

(Phys 4624)

Exercise: Numerical Estimation of a Falling Object with Air Resistance

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1 Problem Statement

The acceleration of a falling object subject to air resistance is described by the following differential equation:

$$\frac{dv}{dt} = g - \frac{c}{m}v \tag{1}$$

where v is velocity, g is acceleration due to gravity, c is the air drag coefficient, and m is mass. The analytical solution for this equation, assuming $v(0) = 0$, is given by:

$$v(t) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right) \tag{2}$$

1.1 Parameters

For this simulation, the following constants are used:

- $g = 9.8$ m/s^2
- $m = 5.5$ kg
- $c = 1$ kg/s
- Time range: $t = 0$ to 20 s
- Step sizes: $\Delta t = 0.1$ s and $\Delta t = 0.05$ s

2 Numerical Methodology

We use the **Euler Method** to estimate the velocity:

$$v_{i+1} = v_i + \left(g - \frac{c}{m}v_i\right)\Delta t \tag{3}$$

3 Numerical Estimation Table

Table 1: Comparison of Euler Method Estimations for $\Delta t = 0.05$ s and $\Delta t = 0.1$ s

Time t (s)	$v(t)$ ($\Delta t = 0.1$)	$v(t)$ ($\Delta t = 0.05$)	**Difference (δ)**
0.5	4.72503	4.70433	0.02070
2.0	16.55692	16.49400	0.06292
5.0	32.36513	32.27431	0.09082
10.0	45.29609	45.22337	0.07272
15.0	50.46246	50.41878	0.04368
19.5	52.39462	52.36969	0.02493

Note: The difference δ peaks mid-simulation and begins to decrease as both solutions converge toward terminal velocity.

3.1 Visualizations

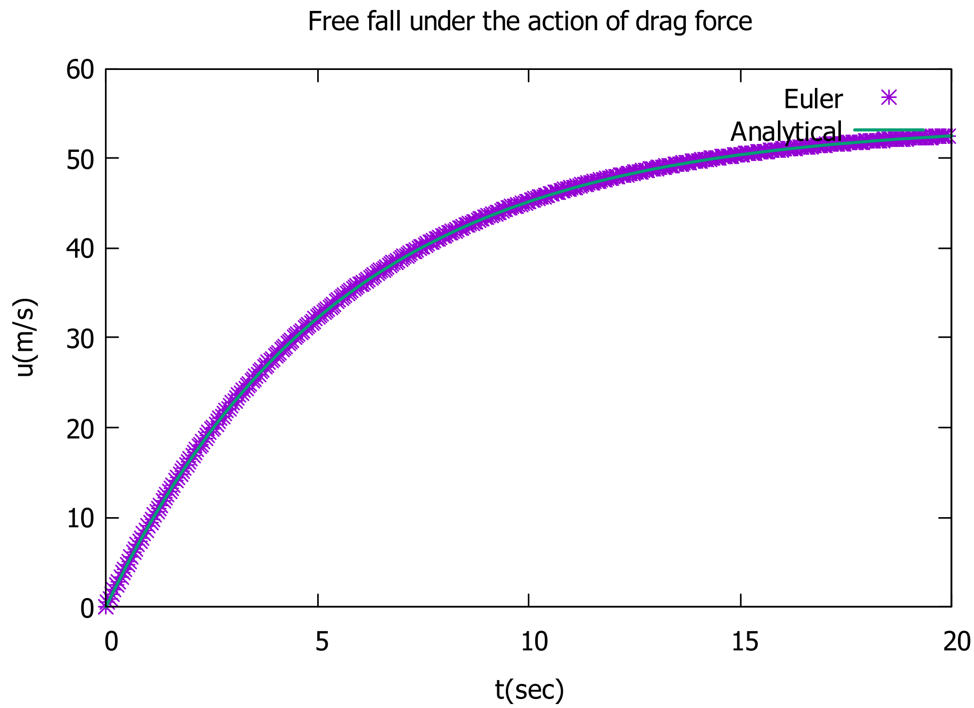

Figure 1: Velocity $v(t)$ as a function of time t for $\Delta t = 0.05$ case. The analytical solution is superimposed in this graph.

4 Comparison and Discussion

In this section, we compare the numerical results with the analytical solution obtained by ignoring air resistance ($v(t) = gt$). The plot for dragged free fall is based on $dt = 0.05s$.

As it is shown, the if resistance were ignored, the particle accelerate uniformly. On the other hand, the one with a air resistance, approaches some define velocity, called terminal velocity v_t. In our case $v_t \approx 52m/s$.

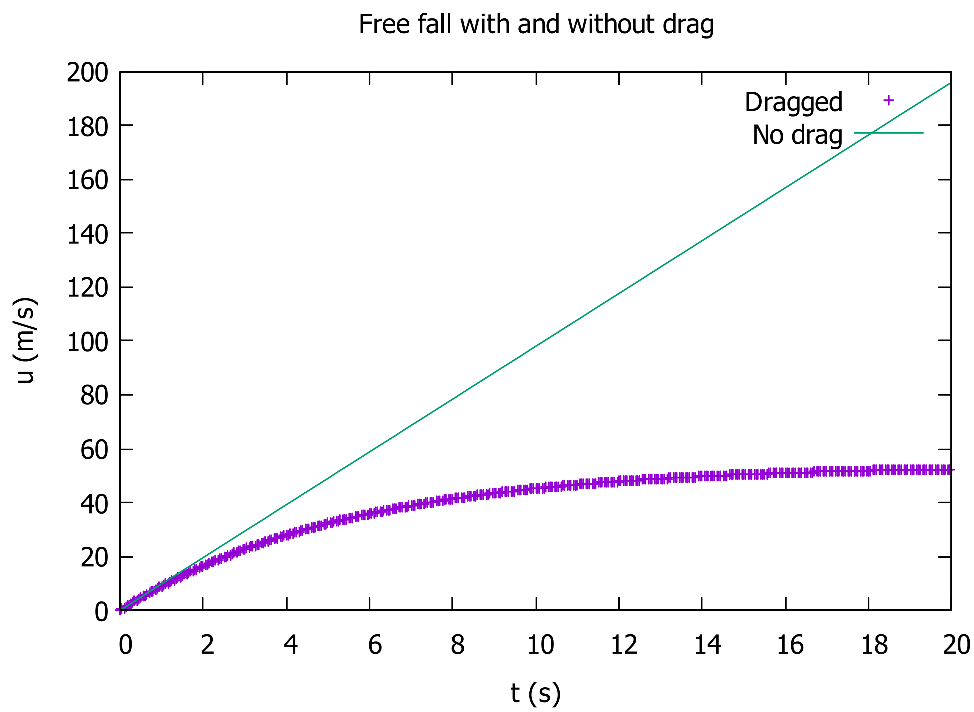

Figure 2: Freely falling body velocity under the influence of gravity. The two curves are for a body experiencing air resistance and with no air resistance.